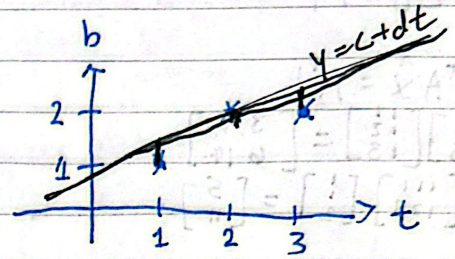
Lecture #16 - Projection Matrices and Least Squares

3/3/26

Least Squares

Fitting by a line

$(1,1), (2,2), (3,2)$



$b = C + Dt$

$t=1 \quad C+D=1$

$t=2 \quad C+2D=2$

$t=3 \quad C+3D=2$

Minimize $\|Ax-b\|^2 = \|e\|^2$

overall error $= e_1^2 + e_2^2 + e_3^2$

↳ want to minimize error

↓

linear regression

the matrix is

$$\underset{A}{\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}} \underset{x}{\begin{bmatrix} C \\ D \end{bmatrix}} = \underset{b}{\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}}$$

No solution, so look for best solution

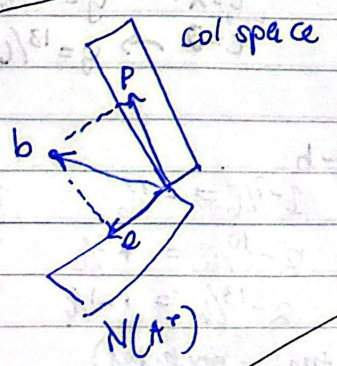
Solve for $A^TA\hat{x} = A^Tb$

Projection Matrix

$P = A(A^TA)^{-1}A^T$

If b in column space $Pb = b$ → projecting vector in col space = itself

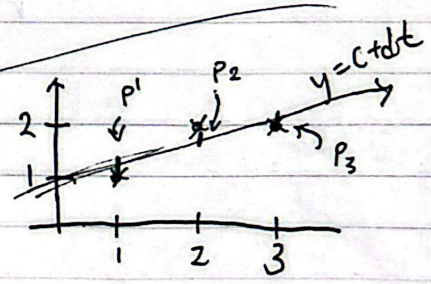
If $b \perp$ column space $Pb = 0$



Pb

$(I-P)b$ → proj onto ⊥ space

$p + e = b$



distance between real point & $y=c+dt$ is e

the points on the line $= p_1 \; p_2 \; p_3$

$$\boxed{A^TA\hat{x} = A^Tb}$$

Find $\hat{x} = \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix}$, p

$$A^T A \hat{x} = A^T b$$

$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$;

$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$

$A^T A$ will always create a square and symmetric matrix

$$\left.\begin{aligned} 3C + 6D &= 5 \\ 6C + 14D &= 11 \end{aligned}\right\} \text{ the normal equations}$$

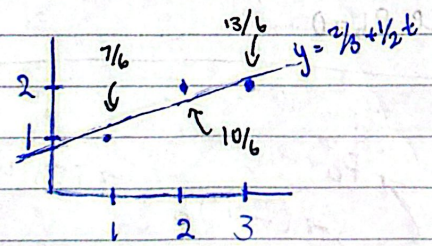
Minimize $\|Ax - b\|^2 = \|e\|^2 = e_1^2 + e_2^2 + e_3^2 = (C+D-1)^2 + (C+2D-2)^2 + (C+3D-2)^2$

$$\begin{aligned} 3C + 6D &= 5 \\ 6C + 14D &= 11 \end{aligned}$$

$$\begin{aligned} 6C + 12D &= 10 \\ -\ 6C + 14D &= 11 \\ \hline -2D &= -1 \\ \underline{D = 1/2} \end{aligned}$$

$$\begin{aligned} 3C + 6(1/2) &= 5 \\ 3C &= 2 \\ \underline{C = 2/3} \end{aligned}$$

Best Line $\rightarrow$ $2/3 + 1/2\, t$



$t=1 \rightarrow y = 7/6$

$t=2 \rightarrow y = 10/6$

$t=3 \rightarrow y = 13/6$

$e = p - b$

$$\left.\begin{aligned} 1 - 7/6 &= -1/6 \\ 2 - 10/6 &= +2/6 \\ 2 - 13/6 &= -1/6 \end{aligned}\right\} \text{ add up to } 0$$

(actual - predicted)

$$\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 10/6 \\ 13/6 \end{bmatrix} + \begin{bmatrix} -1/6 \\ 2/6 \\ -1/6 \end{bmatrix}$$

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$$\begin{cases} A^TA\hat{x} = A^Tb \\ p = A\hat{x} \end{cases}$$

If A has independent columns,
then $\underbrace{A^TA \text{ is invertible}}_{\text{to prove: } x \text{ must be } 0}$.

↳ invertible

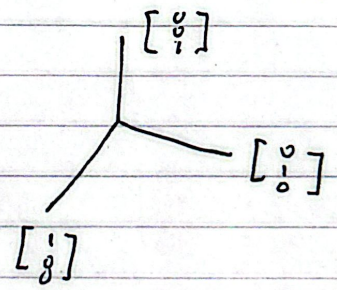
<u>Suppose</u> $A^TAx = 0$

<u>idea</u> $x^TA^TAx = 0$

$$\underbrace{x^T}\underbrace{A^T}\underbrace{Ax} = 0 = (Ax)^T(Ax)$$

$$= \underbrace{Ax = 0}$$

If $Ax = 0$ and A has independent columns, then $x = 0$

$\underline{x = 0}$

For next lecture: Columns are definitely independent if they are $\underbrace{\perp \text{ unit vectors}}_{\text{orthonormal vectors}}$